

NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE

PH1012 QUIZ 1 SEMESTER 1, 2024-2025

14th September 2024, 2:00 pm to 3:30 pm

Name:		Short Q	/ 30
		Q1	/ 10
Matric Number:		Q2	/ 10
		Q3	/ 10
Tutorial Group:		Total	/ 60

Constants and Formula List

Gravitational Acceleration	g	9.8 m/s ²
Mass of Earth	M _e	5.97×10 ²⁴ kg
Gravitational constant	G	6.67×10 ⁻¹¹ N·m ² /kg ²
Avogadro's number	N _A	6.02×10 ²³ molecules/mol
Universal gas constant	R	8.314 J/mol K
Absolute zero	0 K	- 273.15°C
Pressure conversion	1 bar	10 ⁵ N/m ² (or Pa)
	1 atm	1.01 × 10 ⁵ Pa

Basics and Thermal Physics

$$\Delta L = \alpha L_i \Delta T, \quad \Delta V = \beta V_i \Delta T, \quad PV = nRT, \quad Q = mc\Delta T, \quad Q = ml$$

$$T = \frac{P(T) - P(0)}{P(100) - P(0)} \times 100, \quad T = 273.16 \text{ K } \lim_{P_{tr} \rightarrow 0} \frac{P}{P_{tr}},$$

Linear Motion

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_f - \vec{r}_i}{t_f - t_i}, \quad \vec{v}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}, \quad v = \frac{dx}{dt},$$

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}, \quad \vec{a}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2}, \quad a = \frac{dv}{dt}$$

$$v_f = v_i + at, \quad x_f = x_i + \frac{1}{2}(v_i + v_f)t, \quad v_f^2 = v_i^2 + 2a(x_f - x_i), \quad x_f = x_i + v_i t + \frac{1}{2}at^2$$

$$y = (\tan \theta_i)x - \left(\frac{g}{2v_i^2 \cos^2 \theta_i} \right) x^2 \text{ (taking downwards as } -ve, \quad \theta_i \text{ is measured above } +ve \text{ x axis),}$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}, \quad R = \frac{v_i^2 \sin 2\theta_i}{g}$$

$$\Sigma \vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}, \quad \vec{p} = m\vec{v}, \quad f_k = \mu_k N, \quad f_s \leq \mu_s N \quad |F_{sp}| = kx \quad v = \frac{mg}{b} \left(1 - e^{-\frac{b}{m}t} \right)$$

Rotational Motion

$$a_c = \frac{v^2}{r} = r\omega^2, \quad a_t = \frac{d|\vec{v}|}{dt}, \quad s = r\theta, \quad \omega = \frac{d\theta}{dt} = \frac{v}{r}, \quad \alpha = \frac{d\omega}{dt} = \frac{a}{r}; \quad \omega_f = \omega_i + \alpha t, \\ \theta_f = \theta_i + \frac{1}{2}(\omega_i + \omega_f)t, \quad \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i), \quad \theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2$$

Mathematics

$$\tan A = \frac{\sin A}{\cos A}; \quad \sin^2 A + \cos^2 A = 1; \quad \sec^2 A = \frac{1}{\cos^2 A} = \tan^2 A + 1; \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ where } ax^2 + bx + c = 0; \\ \frac{d}{dx}(ax^n) = anx^{n-1}; \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1); \quad \int \frac{dx}{x} = \ln x$$

Instructions:

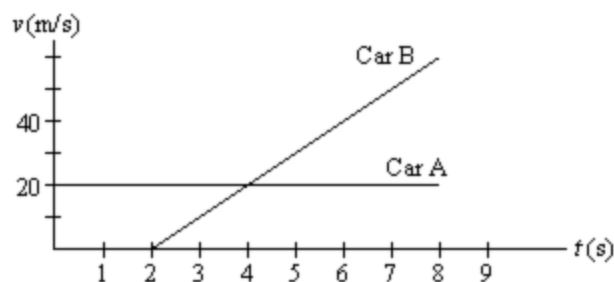
- a) Answer all questions.
- b) Working for all questions must be shown clearly in the space provided.

Short Questions (30 points)

1. [3 points] If a 0.5 GW data center needs 3 months to train an AI model, how long will a 200 MW data center take (in days)? (You can assume that 1 month = 30 days.) Leave your answer to the nearest number of days.

ANS: days

2. [3 points] Driver A is cruising along enjoying the scenery. Driver B starts driving her car at the instant Driver A passes her. Their velocities are shown as functions of time in the graph below. At what instants in time on the graph are Drivers A and B side by side?



- | | | |
|-------------|-------------|-------------|
| A. 0 s, 2 s | B. 0 s, 4 s | C. 0 s, 6 s |
| D. 2 s, 4 s | E. 2 s, 6 s | F. 4 s, 6 s |

ANS:

3. [4 points] A rocket, initially at rest, is fired from the ground vertically with a constant upward acceleration of 4.9 m/s^2 . At an altitude of 0.50 km, the engine of the rocket is turned off. Determine the maximum altitude (in km) of the rocket above the ground.

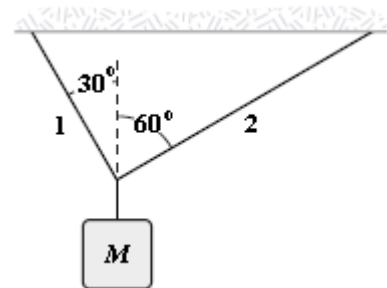
ANS: km

4. [4 points] A playful boy threw a mobile phone at a velocity of 0.95 m/s at 30° above the horizontal. The mobile phone is 1.23 m above the ground when it flew off his hand. Determine the magnitude of the velocity of the mobile phone just before it hits the ground.

ANS:

m / s

5. [4 points] If $M = 5.0 \text{ kg}$, determine the tension in string 2.



ANS:

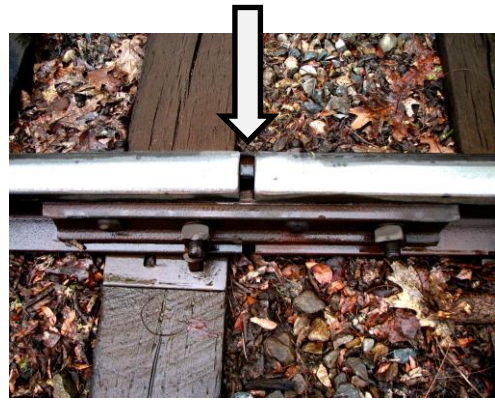
N

6. [2 points] A constant horizontal force is applied to a body that is already moving. As viewed from the top, the force is directed at an angle of 60 degrees to the direction of the body's velocity. What is most likely to happen is that
- A the body will stop moving.
 - B the body will move in the direction of the force.
 - C the body's velocity will increase in magnitude but not change direction.
 - D the body will first stop moving and then move in the direction of the force.
 - E the body will gradually change direction, toward that of the force, while speeding up.

ANS:

7. [3 points] On railway tracks, typically a joint gap is left between rail ends to allow for expansion of the rails in hot weather. The rail tracks are made of steel with a coefficient of linear expansion $\alpha = 12 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. The rail tracks are each 20 m when the temperature is 15°C . In a country where the temperature ranges from -15°C to 55°C , determine the minimum width (in m) of the joint gap needed.

Joint gap on rail track



ANS:

 m

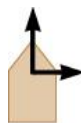
8. [4 points] Determine the minimum area of a flat ice sheet with uniform thickness of 0.50 meter if it is to support a 1200 kg car just above seawater (i.e. the ice sheet is just fully submerged). You are given density of ice $\rho_{\text{ice}} = 920 \text{ kg/m}^3$ and density of seawater $\rho_{\text{sea}} = 1020 \text{ kg/m}^3$.

ANS:

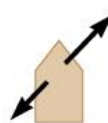
 m^2

9. [3 points] The cutting tool is given two displacements, one of magnitude 4 cm and one of magnitude 3 cm, in each one of five situations (a) through (e) shown below. Rank these situations according to the magnitude of the total displacement of the tool, putting the situation with **the greatest resultant magnitude first**. If the total displacement is the same size in two situations, give those letters equal ranks.

- A $a = c > e > d > b$
 B $c > b > a > e > d$
 C $c = e > a > d > b$
 D $d = b > e > c = a$



a



b



c



d



e

ANS:

Structured Questions (30 points)

Q1. [10 points]

a) As shown in the lecture demonstration, a helium balloon was cooled to about -190°C when liquid nitrogen was poured over it. You can assume that the temperature inside the balloon was uniform. At this point, the volume of the helium balloon has shrunk to about 800 cm^3 and no longer float in air. We can assume that the pressure inside the balloon was always maintained as the same as the outside atmospheric pressure.

- i. Determine the mass of helium gas in the balloon. (Given that the molar mass of He is 4.0 g per mol.)

ANS: g

- iii. If the skin of the balloon had a mass of 2.0 g , determine the temperature (in $^{\circ}\text{C}$) that the balloon would start to float again given that the density of the surrounding air is 1.21 kg/m^3 .

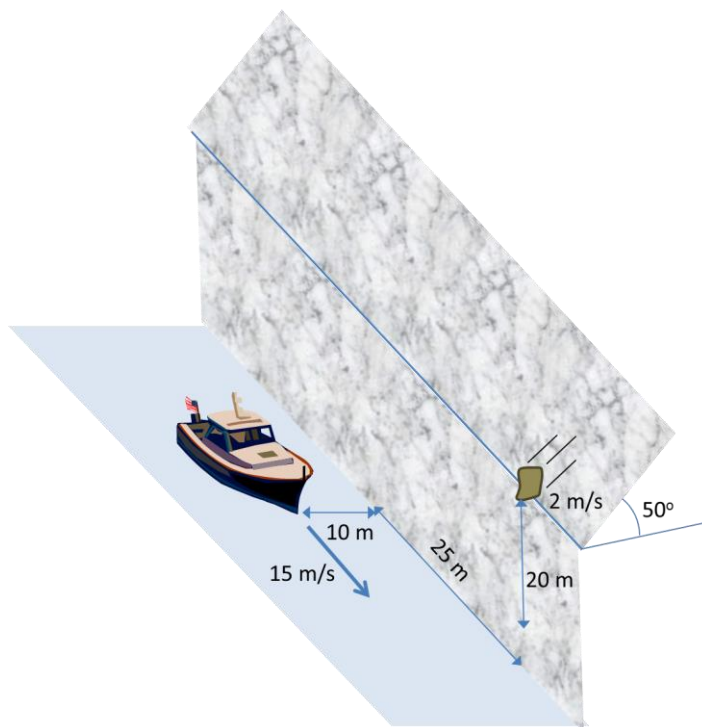
ANS: $^{\circ}\text{C}$

b) You bought a cup of soft drink and you would like to chill it down without diluting it too much. If the volume of the drink is 300 cm^3 at 26°C and you would like your drink to be about 21°C at thermal equilibrium, how many pieces of 1.2 g of ice at -5°C should you add in order to achieve that? (You can assume that your drink has the same density as water, 1.0 g/cm^3 and there is no heat exchanged with the surroundings. The specific heat capacity of ice is $2100\text{ J/kg}^{\circ}\text{C}$, latent heat of fusion of ice is 333 kJ/kg and the specific heat capacity of water is $4186\text{ J/kg}^{\circ}\text{C}$.)

ANS: pieces

Q2. [10 points]

As shown in the figure on the right, a 6 m long and 3 m wide boat moves along a narrow passage way of water next to a granite cliff. The boat is moving at a steady speed of 15 m/s parallel to the cliff at a constant distance of 10 m away (measured from center of the boat shown in the figure). As shown, a small piece of rock slides off the edge of a 50° cliff at 2 m/s when the front tip of the boat is 25 m away from the rock. The rock falls from a height 20 m above the tip of the boat.



- a) We denote the time for the rock to reach the same level as the front tip of the boat as t_{rock} and the time for the boat to reach where the rock has fallen as t_{boat} . By calculating t_{rock} , t_{boat} and doing any other relevant calculations, determine (with a short explanation) if the rock will hit the boat.

ANS: $t_{rock} =$ s

ANS: $t_{boat} =$ s

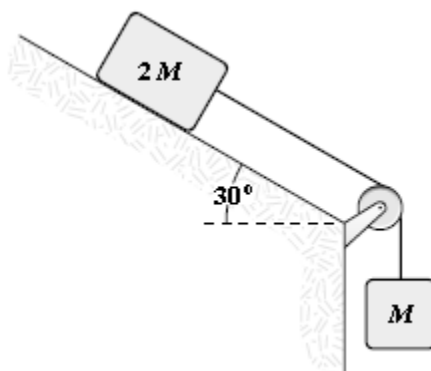
Short explanation:

Will the rock hit the boat? (Write Yes / No in the box. Credit will only be given if above calculations and explanations are correct.)

- b) After the rock slides off at 50° below the horizontal from the cliff, how will the angle of the velocity of the rock be subsequently?
- Increasing to more than 50° below the horizontal.
 - Stay constant at 50° below the horizontal.
 - Reducing to less than 50° below the horizontal.

ANS:

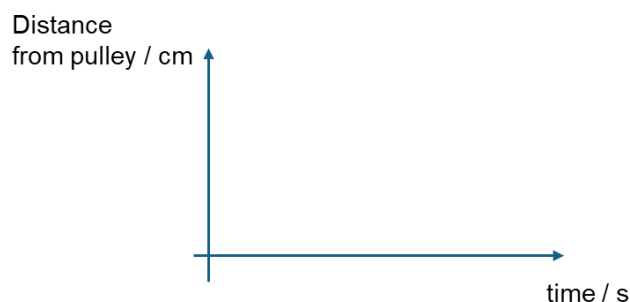
Q3. [10 points] In the figure shown, two blocks are connected by a massless cable round a smooth pulley where friction is negligible. The block on the 30° slope has mass $2M$ and the other suspended block has mass M . The blocks are moving downwards slowly and the coefficient of kinetic friction between the block with mass $2M$ and the incline is 0.20.



- a) Determine the magnitude of the acceleration of the suspended block as it falls (before the block of mass $2M$ moving down the incline hits the pulley).

ANS:

- b) On the axes below, sketch to show how the distance of the suspended block (of mass M) from the pulley changes with time (before the block of mass $2M$ moving down the incline hits the pulley).



- c) Suggest one modification that can be made so that the magnitude of the acceleration (in part a) will be larger.

ANS:

----- The End -----